Sida (Star) Li

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EDUCATION

The University of Chicago — Sept 2024 - Now
Ph.D., Data Science Institute
Mentors: Haifeng Xu, Nikolaos Ignatiadis

The University of Chicago — Sept 2022 - June 2024
Master of Science, Statistics — GPA: 3.97
Thesis Mentor: Daniel Sanz-Alonso

University of California, Berkeley — August 2018 - May 2022
Bachelor of Arts, Statistics & Computer Science — GPA: 3.95
Statistics Department Citation (Valedictorian) Winner

RESEARCH INTERESTS

Reinforcement learning for LLM agents (environments, rewards, and training infrastructure), evaluation of language models, and prediction-powered inference: combining a small amount of reliable data with a large amount of cheap ML predictions, using empirical Bayes ideas.

PREPRINTS

Chenghao Yang, **Sida Li**, Ari Holzman. "LLM Probability Concentration: How Alignment Shrinks the Generative Horizon." **In Submission.** [Preprint]

PAPERS & REPORTS

Wanyi Lin, **Sida Li**, Junmin Guan, Nikolaos Ignatiadis. "Empirical Bayes Rebiasing." **Neural Information Processing Systems (NeurIPS) 2026.** [PDF]

Oussama Boussif, Mohammed Mahfoud, Younesse Kaddar, Moksh Jain, **Sida Li**, Damiano Fornasiere, Xiaoyin Chen, Yoshua Bengio, Nikolay Malkin. "Bayesian Symbolic Regression with Entropic Reinforcement Learning." **Conference on Uncertainty in Artificial Intelligence (UAI) 2026.** [OpenReview]

Siyang Wu*, Honglin Bao*, Sida Li*, Ari Holzman, James Evans. "Mapping Overlaps in Benchmarks through Perplexity in the Wild." **International Conference on Learning Representations (ICLR) 2026.** [Arxiv]

Qingchuan Yang*, Simon Mahns*, Sida Li*, Anri Gu*, Jibang Wu, Haifeng Xu. "LLM-as-a-Prophet: Understanding Predictive Intelligence with Prophet Arena." **International Conference on Learning Representations (ICLR) 2026**. [Blogpost] [Platform] [Arxiv]

Jingyang Yi, Jiazheng Wang, **Sida Li**. "ShorterBetter: Guiding Reasoning Models to Find Optimal Inference Length for Efficient Reasoning." **Neural Information Processing Systems (NeurIPS) 2025**. [ArXiv]

Sida Li, Nikolaos Ignatiadis. "Prediction-Powered Adaptive Shrinkage Estimation." **International Conference on Machine Learning (ICML) 2025**. [ArXiv]

Sebastian Musslick, Benjamin Andrew, Chad C. Williams, **Sida Li**, Ioana Marinescu, Marina Dubova, George T. Dang, Younes Strittmatter, John G. Holland. "AutoRA: Automated Research Assistant for Closed-Loop Empirical Research." **Journal of Open Source Software, 2024.** [Link]

Sida Li, Ioana Marinescu, Sebastian Musslick. "GFN-SR: Symbolic Regression with Generative Flow Networks." **NeurIPS 2023 AI4Science Workshop**. [ArXiv] [Poster]

Sebastian Musslick, Joshua Hewson, Ben Andrew, **Sida Li**, George Dang, John Gerrard Holland. "Evaluating Computational Discovery in the Behavioral and Brain Sciences." **AAAI 2023 Spring Symposium Series, Computational Approaches to Scientific Discovery**. [Talk Abstract]

Michael Estrada, **Sida Li**, Xiangyu Cai. "Feedback Linearization of Car Dynamics for Racing via Reinforcement Learning." **Preprint, 2021**. [Link]

OPEN SOURCE

rLLM (Agentic Reinforcement Learning) [Recent Blogpost] [Github 5.2K stars]
Core contributor of the project. Maintaining the rLLM ecosystem (social media, developer Slack channel, etc.) and building the agent orchestration layer that allows post-training any agentic program using any algorithm.

Automated Research Assistant (AutoRA) [Link]
An open-source framework for automating multiple stages of the empirical research process, including model discovery, experimental design, data collection, and documentation for open science.

THESIS

Beyond Vanilla Metropolis-Adjusted Langevin Dynamics. *Mentored by Prof. Daniel Sanz-Alonso.*

INVITED TALK

Prediction-Powered Adaptive Shrinkage Estimation. *International Seminar on Selective Inference.* April, 2025. [Link]

WORK EXPERIENCE

Member of Technical Staff Intern, Stealth Frontier Data Lab *Fall 2026 - Present*
Building synthetic environments for training LLM agents with reinforcement learning.

Research Scientist Intern, Snowflake, Bellevue, WA *March - Sept 2026*
Worked within the post-training team to build agentic systems that quickly adapt to personalization needs with few-shot demonstrations.

Software Engineer Intern, Duolingo, Pittsburgh, PA *May - August 2021*
Implemented internal tools in the ETL data pipeline that support efficient querying and computation on key metrics (e.g. daily bookings, active users); revised the A/B testing framework by enabling auto-correction in confidence intervals for ad-hoc metrics.

Data Consulting Intern, Concha Inc., Berkeley, CA *January - May 2020*
Worked on predicting customers' hearing loss curves based on response data from online testing. Applied and evaluated existing machine learning methods such as regression trees and RNNs for the prediction tasks.

RESEARCH EXPERIENCE

The University of Chicago Statistics Department, UChicago, IL
Mentor: Nikolaos Ignatiadis *April 2024 - Present*
Researching empirical Bayes mean estimation problems under semi-supervised assumptions. Extending the prediction-powered inference (PPI) framework to a compound decision setting and building a new empirical Bayes estimator that (1) utilizes massive unlabeled data with PPI de-biasing and (2) enjoys risk guarantees comparable to the full-Bayesian oracle estimator.

UChicago Master Thesis, UChicago, IL
Mentor: Daniel Sanz-Alonso *June 2023 - April 2024*
Worked on accelerating and generalizing Langevin Monte Carlo (LMC) methods for sampling. Experimented and verified how adding a curl matrix into the Langevin SDE accelerates convergence in various statistical models. Implemented and benchmarked a new Fisher-information based LMC method that outperformed traditional counterparts in various metrics.

Autonomous Empirical Research Group, Brown University, RI
Mentor: Sebastian Musslick *March 2022 - Present*
Researching symbolic regression (SR), the ML problem of searching for the best-fitting expression for a given dataset. Developed a hierarchical Bayesian framework for SR and a corresponding inference algorithm to sample from the posterior. Pioneered the design of a new SR method based on Generative Flow Networks (GFlowNets) that achieves SOTA performance in noisy settings.

Sandrine Dudoit Lab, UC Berkeley, CA
Mentor: Hector Roux de Bezieux and Koen Van den Berge *January - May 2020*
Participated through the Undergraduate Research Apprentice (URAP) program. Investigated how initialization affects unsupervised dimensionality reduction methods such as UMAP and t-SNE for scRNAseq data, with an emphasis on the preservation of global structures in low dimensional space.

ACADEMIC SERVICE Reviewer for ICLR 2026, AISTATS 2026

AWARDS

MMLS 2025, Best Poster Runner Up	SU25
UChicago M.S. Stat Scholarship (25% tuition remission)	FA22, FA23
UC Berkeley Statistics Department Citation	FA22
UC Berkeley Dean's Honors List (top 10% GPA)	SP19, FA19, SP20, SP21
Upsilon Pi Epsilon (top one third of CS majors)	FA19, SP20, FA20, SP21

TEACHING

At UChicago

DATA 37711 Foundations of ML & AI (Part I) (TA)	Fall 2025
DATA 21200 Mathematical Methods for Data Science II (TA)	Spring 2025

At UC Berkeley

CS 198-097 Robot Autonomous Racing DeCal (Head Instructor)	Fall 2021
CS 198-097 Robot Autonomous Racing DeCal (Instructor)	Fall 2020
STAT 134 Probability Theory (Tutor)	Spring 2020
STAT 134 Probability Theory (Tutor)	Fall 2019
MATH 32 Precalculus (Tutor)	Summer 2019

SKILLS

Languages: English, Mandarin, Cantonese
Programming: Python, R, Java, Javascript, Ruby, C++, LaTeX
Frameworks: PyTorch, Pandas, NumPy, Scikit-learn